A Diamond Subspace Lattice Whose Complement Cannot Be Isometric

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Status

This packet gives a candidate counterexample to the isometric-complement variant highlighted in Section 3.3 of Bracic–Kandic [1]. It shows that even when $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ is complemented in $\mathrm{Col}(\mathfrak{L})$, one cannot in general choose the complement to consist of isometric operators.

The example is only about the isometric representative question. It does not settle the harder complement problem for Volterra nests on $L^p[0,1]$, $1 < p < \infty$.

Source Question

In Section 3.3, after defining complements of $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ in $\mathrm{Col}(\mathfrak{L})$, the source says:

> **3.3. A complement of $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ in $\mathrm{Col}(\mathfrak{L})$.** Let $\mathfrak{L} \subseteq \mathfrak{C}(\mathscr{X})$ be a subspace lattice. By Theorem 3.7, $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ is a normal subgroup of $\mathrm{Col}(\mathfrak{L})$. It is an interesting question whether $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ is a complemented subgroup of $\mathrm{Col}(\mathfrak{L})$, that is, whether there exists a subgroup $\mathcal{O}(\mathfrak{L}) \subseteq \mathrm{Col}(\mathfrak{L})$ called a complement of $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ in $\mathrm{Col}(\mathfrak{L})$ such that $\mathrm{Col}(\mathfrak{L}) = \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)\mathcal{O}(\mathfrak{L})$ and $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big) \cap \mathcal{O}(\mathfrak{L}) = \{I\}$. If this happens, then $\mathrm{Col}(\mathfrak{L})$ is an inner semidirect product of $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ and $\mathcal{O}(\mathfrak{L})$, which we write as $\mathrm{Col}(\mathfrak{L}) = \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big) \rtimes \mathcal{O}(\mathfrak{L})$. Moreover, since $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ is a normal subgroup of $\mathrm{Col}(\mathfrak{L})$, we have $\mathrm{Col}(\mathfrak{L}) = \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)\mathcal{O}(\mathfrak{L}) = \mathcal{O}(\mathfrak{L})\,\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$. Indeed, if $S = AT$, with $A \in \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ and $T \in \mathcal{O}(\mathfrak{L})$, then $S = TB$, where $B = T^{-1}AT \in \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$.
>
> If a complement of $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ in $\mathrm{Col}(\mathfrak{L})$ exists, it does not need to be unique. Indeed, if $\mathcal{O}(\mathfrak{L})$ is a complement of $\mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ and $\mathcal{O}(\mathfrak{L})$ is not a normal subgroup of $\mathrm{Col}(\mathfrak{L})$, then there exists $C \in \mathrm{Col}(\mathfrak{L})$ such that $C\mathcal{O}(\mathfrak{L})C^{-1} \neq \mathcal{O}(\mathfrak{L})$. It is not hard to see that $C\mathcal{O}(\mathfrak{L})C^{-1}$ is a complement of $\mathrm{Grp}(\mathfrak{L})$, as well. Although complements of $\mathrm{Grp}(\mathfrak{L})$ in $\mathrm{Col}(\mathfrak{L})$ are possibly not unique, they are all isomorphic to the quotient group $\mathrm{Col}(\mathfrak{L})/\,\mathrm{Grp}(\mathfrak{L})$ via the natural group isomorphism $\mathcal{O}(\mathfrak{L}) \to \mathrm{Col}(\mathfrak{L})/\,\mathrm{Grp}(\mathfrak{L})$ given by $T \mapsto T\,\mathrm{Grp}(\mathfrak{L})$. Of special interest is the question of whether there exists a complement consisting of isometric operators. Notice that this trivially holds if $\mathrm{Col}(\mathfrak{L}) = \mathrm{Grp}\big(\mathrm{Alg}(\mathfrak{L})\big)$ in which case we have $\mathcal{O}(\mathfrak{L}) = \{I\}$. The following simple proposition, whose proof is

The exact target addressed here is the sentence asking whether there exists a complement consisting of isometric operators.

Definitions Used

For a subspace lattice $\mathfrak{L}$ on a complex Banach space X, $\mathrm{Col}(\mathfrak{L})$ is the group of invertible operators S such that

$$M \in \mathfrak{L} \iff SM \in \mathfrak{L}.$$

The subgroup $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ consists of the invertible operators that leave each member of $\mathfrak{L}$ invariant. A complement is a subgroup $\mathcal{O}(\mathfrak{L}) \subseteq \mathrm{Col}(\mathfrak{L})$ such that

$$\mathrm{Col}(\mathfrak{L}) = \mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))\, \mathcal{O}(\mathfrak{L}), \qquad \mathrm{Grp}(\mathrm{Alg}(\mathfrak{L})) \cap \mathcal{O}(\mathfrak{L}) = \{I\}.$$

Proof Intuition

The smallest nontrivial quotient is the diamond lattice, where the quotient $\mathrm{Col}(\mathfrak{L})/\,\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ has order two and records whether a collineation preserves or swaps the two atoms. Algebraically, a swapping representative always exists when the two atoms are linearly isomorphic. But an isometric representative would impose symmetry of the norm between the two coordinate axes. We choose a complex norm on $\mathbb{C}^2$ whose unit ball is deliberately tilted by the functional $z + w/2$. This keeps the linear diamond intact while destroying every isometric axis swap.

Theorem 1. *There exists a two-dimensional complex Banach space X and a subspace lattice $\mathfrak{L} \subseteq \mathfrak{C}(X)$ such that $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ is complemented in $\mathrm{Col}(\mathfrak{L})$, but no complement consists of isometric operators.*

Proof. Let $X = \mathbb{C}^2$ with norm

$$\|(z, w)\| = \max\{|z|,\ |w|,\ |z + \tfrac{1}{2}w|\}.$$

This is a norm because the three linear functionals z, w, and $z + \frac{1}{2}w$ separate points. Let

$$M = \mathbb{C}(1, 0), \qquad N = \mathbb{C}(0, 1), \qquad \mathfrak{L} = \{\{0\}, M, N, X\}.$$

An invertible operator is a collineation of $\mathfrak{L}$ precisely when it permutes the two one-dimensional atoms M and N. Therefore $\mathrm{Col}(\mathfrak{L})$ is the group of invertible monomial 2×2 matrices relative to $X = M \oplus N$. The subgroup $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ is the diagonal subgroup, namely the invertible operators preserving M and N separately. Hence the quotient is the two-element group S_2, and

$$W(z, w) = (w, z)$$

gives an algebraic complement $\{I, W\}$. Thus $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$ is complemented in $\mathrm{Col}(\mathfrak{L})$.

We now show that no complement can consist of isometries. If such a complement existed, its nontrivial element U would represent the nontrivial coset in $\mathrm{Col}(\mathfrak{L})/\,\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$. Thus U would have to swap M and N, so there would be nonzero scalars $a, b \in \mathbb{C}$ with

$$U(1, 0) = a(0, 1), \qquad U(0, 1) = b(1, 0).$$

Since $\|(1, 0)\| = \|(0, 1)\| = 1$, isometry of U forces $|a| = |b| = 1$.

Set $x = (1, \frac{1}{8})$. Then

$$\|x\| = \max\left\{1, \frac{1}{8}, \left|1 + \frac{1}{16}\right|\right\} = \frac{17}{16}.$$

On the other hand,

$$Ux = \left(\frac{b}{8}, a\right),$$

and hence

$$\|Ux\| = \max\left\{\frac{1}{8},\ 1,\ \left|\frac{b}{8} + \frac{a}{2}\right|\right\} \le \max\left\{\frac{1}{8},\ 1,\ \frac{1}{8} + \frac{1}{2}\right\} = 1.$$

This contradicts the assumption that U is an isometry. Therefore the nontrivial coset has no isometric representative, and no isometric complement exists. □

Verification Notes

All operators involved are finite-dimensional. The collineation group of the diamond lattice is exactly the monomial group because an invertible linear map must permute the two atoms M and N. The diagonal subgroup is exactly $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{L}))$. The no-isometry obstruction uses only the single test vector $(1, 1/8)$, so there is no hidden compactness or approximation issue.

Limitations

This packet does not answer whether $\mathrm{Grp}(\mathrm{Alg}(\mathfrak{N}))$ is complemented in $\mathrm{Col}(\mathfrak{N})$ for the Volterra nest $\mathfrak{N} \subseteq \mathfrak{C}(L^p[0,1])$, $1 < p < \infty$. It also does not rule out isometric complements in special symmetric settings, such as Hilbert-space examples where the quotient automorphisms have unitary representatives.

Novelty Check

The local run indexes were searched for arXiv:2508.14603, the source title, subspace-lattice and collineation keywords, complemented subgroup, isometric complement, isometric operators, and diamond-lattice variants. No exact prior packet for this question was present.

External phrase searches on June 28, 2026 used combinations of the phrases isometric complement, subspace lattice, collineations, and Grp(Alg). They did not locate a published answer. Since the source paper is recent, the novelty check should be regarded as bounded rather than exhaustive.

Human Review Recommendation

Check that the page-12 sentence is meant in the literal generality of arbitrary complex Banach spaces and subspace lattices. Under that reading the example is a complete negative answer to the isometric-complement variant. If the intended question was restricted to the later Volterra nests, this packet should be read as a small finite-dimensional scope counterexample.

References

[1] J. Bracic and M. Kandic, *Similarities of subspace lattices in Banach spaces*, arXiv:2508.14603.